

LagaTour — Backend Changes & Technical Documentation

Summary of Code Contributions, Database Architecture & API Endpoints

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Commits Created: `66af138`, `57a6f36`

Project Repository: LinkonPenaru/LagaTour

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1. Executive Summary

This update implements the entire **Real-time Chat & Messaging System** (Step 1, Step 2, and Step 3) for the LagaTour platform, along with automated database migrations, Socket.io WebSocket architecture, and complete conflict resolution with the newly pulled Places & Divisions modules.

2. File-by-File Breakdown of Changes

server/src/services/socketService.js

NEW FILE

Purpose: Implements the Socket.io real-time WebSocket server engine.

- **Connection Handling:** Initializes WebSocket server on HTTP server with CORS configuration.
- **Room Subscriptions:** Listens for `join_chat` and `leave_chat` events.
- **Live Typing Indicators:** Broadcasts `user_typing` and `user_stop_typing` events to room members.
- **Real-time Messaging:** Broadcasts `receive_message` events instantly across active chat rooms.
- **API Hook:** Exports `emitToConversation(convId, event, data)` allowing REST controllers to trigger instant WebSocket updates.

server/src/controllers/chatController.js

NEW FILE

Purpose: Complete business logic and MySQL query handler for messaging.

- `getConversations` : Fetches a traveler's active 1-on-1 DMs and group chats with unread badge count and last message preview.
- `getMessages` : Fetches full message history with pagination (`limit` , `offset`) and updates `last_read_at` .
- `getOrCreateDM` : Checks if a direct 1-on-1 chat exists between two users or creates a new conversation.
- `createGroupChat` : Inserts a group conversation with selected members and posts an automated welcome notice.
- `sendMessage` : Saves messages to MySQL, updates timestamps, awards +2 League Points, and broadcasts via Socket.io.
- `searchChatUsers` : Live database search for travelers by username, name, or bio.

server/src/routes/chatRoutes.js

NEW FILE

Purpose: Express router defining all REST endpoints for Chat & Messaging.

- Mapped `GET /` , `POST /direct` , `POST /group` , `GET /:id/messages` , `POST /:id/messages` , and `GET /users` .

server/src/config/db.js

MODIFIED

Purpose: Database connection pool, schema initialization, and seed data.

- **New Tables:** Added auto-creation for `conversations` , `conversation_members` , and `messages` with `CASCADE` foreign keys.
- **Preserved Modules:** Preserved newly pulled `divisions` , `districts` , `places` , `place_images` , `myplaces` , and `place_ratings` tables.
- **Query Engine:** Upgraded `query()` to use `p.query()` for compatibility with MySQL dynamic clauses.
- **Seed Data:** Added initial sample chats (2 DMs + 1 Group Chat) and messages for demo testing.

server/src/index.js

MODIFIED

Purpose: Main entry point for the Node.js backend server.

- **HTTP Server:** Wrapped Express app in Node's `http.createServer(app)` to support dual HTTP + WebSocket protocols.
- **Socket.io Initializer:** Invoked `initSocket(server)` on startup.
- **Route Mounts:** Mounted both `app.use("/api/chats", chatRoutes)` and `app.use("/api/places", placeRoutes)` .

server/package.json & package-lock.json

MODIFIED

Purpose: Added `socket.io` (v4.x) library for WebSocket communications.

3. REST API Endpoints Reference

Method	Endpoint	Description
GET	/api/chats?userId=:id	Returns all active direct and group conversations for the user with last messages and unread counts.
GET	/api/chats/:id/messages?userId=:id	Retrieves paginated message history for a conversation and marks as read (updates last_read_at).
POST	/api/chats/direct	Finds an existing 1-on-1 chat or creates a new direct conversation between two travelers.
POST	/api/chats/group	Creates a new multi-member group chat with admin assignment and system welcome notice.
POST	/api/chats/:id/messages	Saves a message to MySQL, awards points, and emits instant Socket.io broadcast to room.
GET	/api/chats/users?q=:query	Live search for travelers by username, name, or bio to start new chats with.

4. Database Schema Reference (MySQL: lagatour_db)

Table Name	Key Columns & Constraints	Description & Relationship
conversations	conversation_id (PK) type ('direct' 'group') title , avatar_url created_by (FK → users)	Stores 1-on-1 direct conversations and multi-user group chats. Cascades with members and messages.
conversation_members	member_id (PK) conversation_id (FK → conversations) user_id (FK → users) role ('admin' 'member') last_read_at (DATETIME)	Maps participants to conversations and tracks individual last-read timestamps for unread count calculation.
messages	message_id (PK) conversation_id (FK → conversations) sender_id (FK → users) message_text , media_url message_type ('text' 'image' 'video' 'system')	Stores persistent message history. Indexed on (conversation_id, created_at) for fast message stream lookups.

5. Merge Conflict Resolution Summary

Merged Remote Branch (Commit 57a6f36):

- Places API: Preserved all routes in server/src/routes/placeRoutes.js and controller in server/src/controllers/placeController.js .
- Geographic Database: Preserved tables for divisions (8 Divisions), districts (64 Districts), places , place_images , myplaces , and place_ratings in server/src/config/db.js .
- Clean Working Tree: Pushed clean merge commit 57a6f36 to origin/main on GitHub with zero broken references.